## Building sites

The different stages in the construction of a large building must always take place in a certain order, starting with the preparation of the site. Materials and machinery must arrive just when they are needed: if they are too early, the site may get too crowded; if they are too late, the building work may be delayed.

### Site clearance and excavation

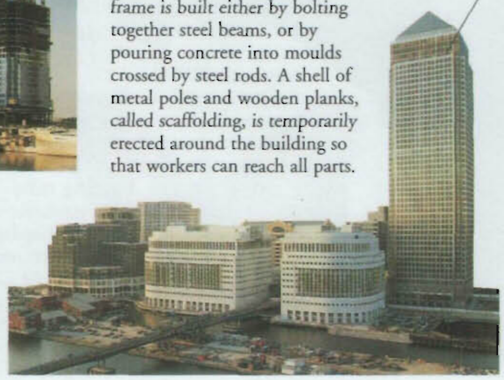
The building site must first be cleared, which may involve demolishing other buildings, removing vegetation, and levelling the site. Holes are excavated (dug) for the foundations and basement.

### Foundation laying

The next stage is to build the foundations. This involves driving steel beams, called piles, into the ground, or pouring liquid concrete into a deep pit to form a solid base that will support the building.

### Frame building

The building's frame soon rises from the foundations. The frame is built either by bolting together steel beams, or by pouring concrete into moulds crossed by steel rods. A shell of metal poles and wooden planks, called scaffolding, is temporarily erected around the building so that workers can reach all parts.




### Completion

With the frame in place, work starts on the floors, walls, and roof. Services such as water and waste pipes, heating and air-conditioning ducts, and electricity and telephone cables are installed on each storey. Finally, the windows are inserted, and the interior is decorated.

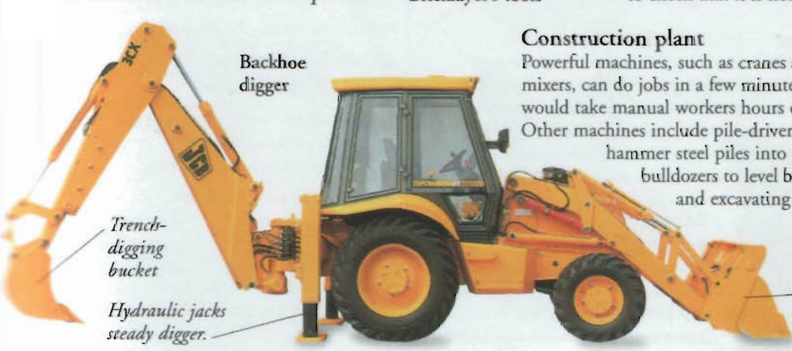
## Equipment

Some of the tasks on a building site, such as plastering a wall or laying bricks, are done by tradespeople using hand tools. Other tasks, such as erecting the building's frame or lifting heavy objects, may require large, specialized machines. Together, these machines are known as construction plant.


**Bricklayer's tools**

### Hand tools

Each tradesperson involved in building and construction uses special tools. A bricklayer, for example, uses a trowel to spread mortar on to bricks, a plumbline to ensure that a wall is vertical, and a spirit level and a set square to check that it is horizontal.




**Backhoe digger**

### Construction plant

Powerful machines, such as cranes and cement mixers, can do jobs in a few minutes that would take manual workers hours or even days. Other machines include pile-drivers to hammer steel piles into the ground, bulldozers to level building sites, and excavating diggers.

## Building materials

Some building materials, such as steel, concrete, and bricks, are structural – that is, they make up the basic structure of the building. Other materials, such as ceramics and glass, are mainly decorative. Traditional materials, such as stone and wood, have been used for many centuries and are often found locally.


**Building site materials**

### Concrete and steel

Most modern buildings contain concrete, steel, or a combination of both. Concrete is a mixture of cement, water, and small stones (called aggregate) that hardens like rock when it sets. Steel is iron that contains a tiny amount of carbon. Concrete strengthened by steel rods is called reinforced concrete.

**Types of concrete**

### Wood

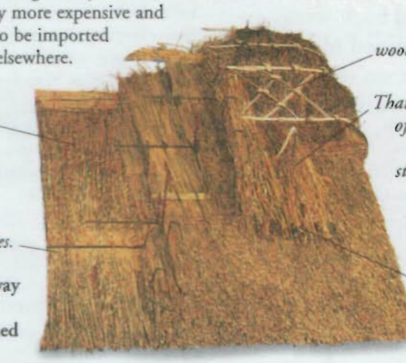
Some houses have floors made of wooden planks and wooden beams for roof trusses. Scaffolding may have walkways of wooden planks.

### Bricks

Blocks of hardened clay, called bricks, are laid in rows and joined together with mortar – a mixture of cement and sand.

### Local materials

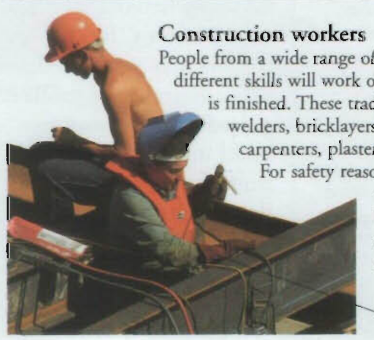
Many buildings throughout the world are built from materials that occur naturally in the surrounding area. These local materials may include straw, mud, stone, wood, and even animal dung. They can do just as good a job as modern manufactured materials, which are usually more expensive and have to be imported from elsewhere.




**Cutaway of a thatched roof**

### Construction workers

People from a wide range of trades with many different skills will work on a building before it is finished. These tradespeople include welders, bricklayers, electricians, carpenters, plasterers, and plumbers. For safety reasons, construction workers often wear hard hats and other protective clothing, such as goggles.



*Welder wearing safety visor and gloves*

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